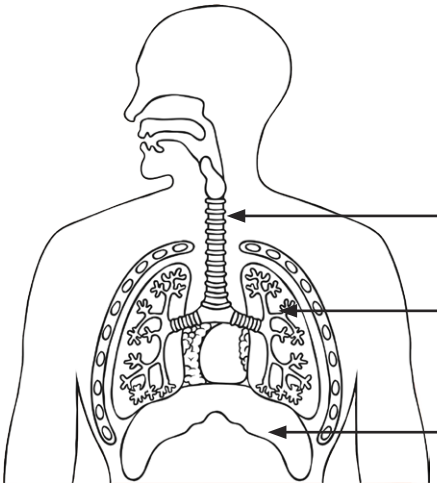
Name:	Class:	Date given:
		Date due in:

Gas Exchange **System**

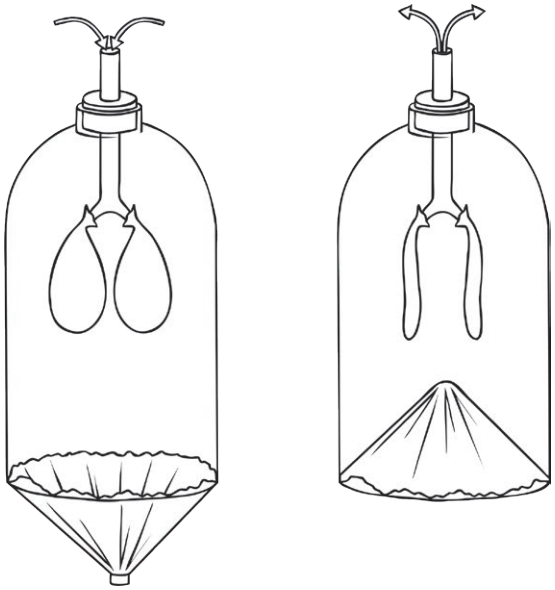


P is _____

Q is _____

R is _____

1. Label P, Q and R on the diagram above. [3]



2. The diagram above shows a piece of apparatus which can be used to model the mechanism of the gas exchange system. Explain how the parts of the gas exchange system control breathing. [4]

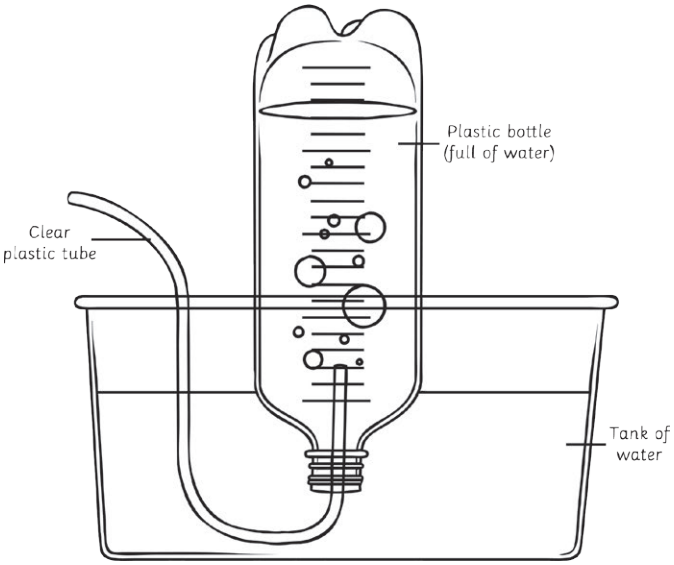
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3. Use the diagram above to explain how the volume of a person's lungs can be measured. [3]

4. Explain how smoking can damage the gas exchange system, and the effect this has on a person. [4]

5. For each statement below, write whether it is **true** or **false**. [4]

   When you exercise you breathe faster _____

   When you exercise you need less energy _____

   After exercise, breathing returns to normal immediately _____

   During exercise the breathing rate is decreased _____

6. What is asthma? [2]

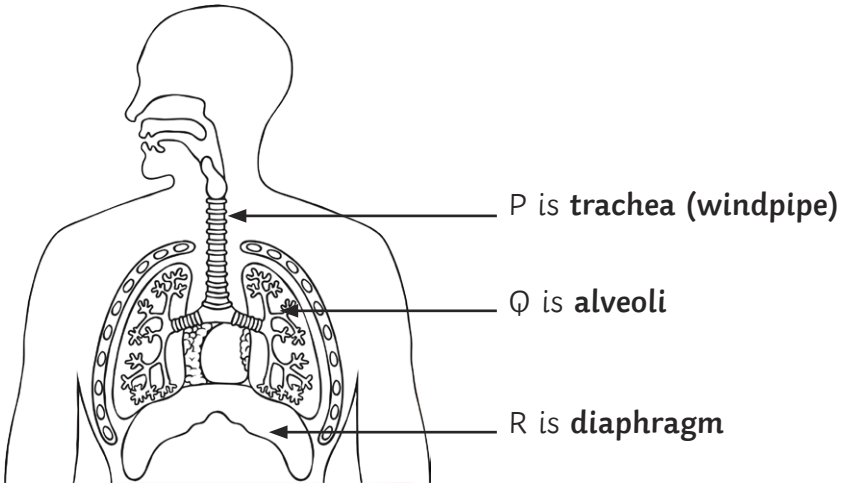
Learning Outcomes (tick if achieved)

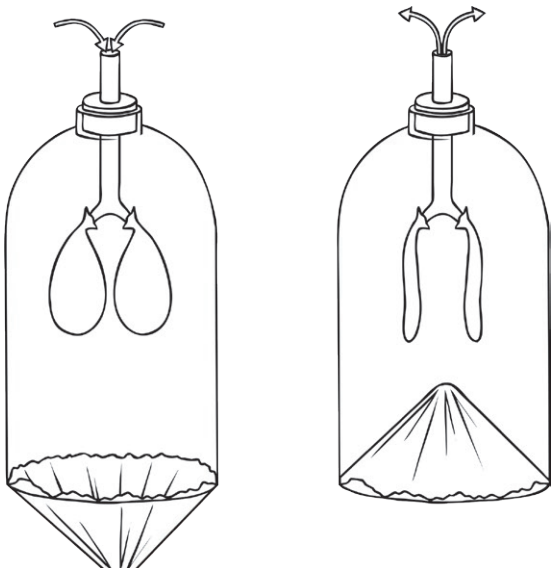
Q1,2	I can describe the structure and function of the gas exchange system	
Q3	I can describe how lung volume is measured	
Q4	I can explain how smoking affects the gas exchange system	
Q5	I can explain how exercise affects the gas exchange system	
Q6	I know what asthma is	

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Gas Exchange System **Answers**

1.





2. The diagram above shows a piece of apparatus which can be used to model the mechanism of the gas exchange system. Explain how the parts of the gas exchange system control breathing. [4]

Max 4 from:

- **The diagram contracts / moves down**
- **Lungs get bigger / expand / capacity increases**
- **Inhale / air drawn into lungs / breathing in**
- **The diaphragm relaxes / moves up**
- **Lungs get smaller / compressed / capacity decreases**
- **Exhale / air pushed out / breathing out**

3. Use the diagram above to explain how the volume of a person's lungs can be measured. [3]

   Max 3 from:

   - **Air breathed out / exhaled through tube**
   - **Air forces water from the bottle / water is displaced**
   - **Air bubble / amount of air measured by volume**
   - **After deep breath the air in bottle is equal to capacity of lungs**

4. Explain how smoking can damage the gas exchange system, and the effect this has on a person. [4]

   Max 4 from:

   - **Harder to breathe / shortness of breath**
   - **Causes coughing / wheezing**
   - **Tar builds up**
   - **Blocked airways**
   - **Alveoli damaged**
   - **Surface area for gas exchange reduced**
   - **Less oxygen into body**
   - **Heart works harder to compensate / can lead to heart attack**
   - **Can cause lung cancer**

5. For each statement below, write whether it is **true** or **false**. [4]

When you exercise you breathe faster	**True**
When you exercise you need less energy	**False**
After exercise, breathing returns to normal immediately	**False**
During exercise the breathing rate is decreased	**False**

6. What is asthma? [2]

   A condition which makes breathing harder / difficulty with breathing

   Usually an allergic reaction (to e.g. pet hair, dust)